Tanmay Bhuskute

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SUMMARY

Software Engineer and M.S. Computer Science candidate (Dec 2026) with production engineering experience building scalable backend systems, full-stack applications, and cloud deployments using Python, Java, JavaScript, React, AWS, and Kubernetes. Current research applies agentic LLM tool-calling and prompt engineering to Android automation.

EDUCATION

Master of Science in Computer Science Aug 2024 - Dec 2026
University of Cincinnati
Relevant Coursework: Advanced Algorithms, Deep Learning, Intelligent Data Analysis, Database Theory.

Bachelor of Engineering in Computer Engineering Aug 2018 - Jul 2022
AISSMS College of Engineering
Relevant Coursework: Data Mining, Data Analytics, Machine Learning, Object-Oriented Programming.

TECHNICAL SKILLS

Programming: Java, Python, JavaScript, TypeScript, C/C++, SQL, Go.

Web & Backend: React, Node.js, Django, Flask, FastAPI, REST APIs, Object-Oriented Design.

Databases: PostgreSQL, MySQL, MongoDB, Redis, Data Structures & Algorithms.

Developer Tools: Git, GitLab CI/CD, Jenkins, AWS, Docker, Kubernetes, ArgoCD, Helm, Terraform, Linux, Bash, Claude Code, GitHub Copilot.

AI & Data: Gemini, Vertex AI, Agentic AI, Prompt Engineering, LLM Tool Calling, PyTorch, TensorFlow, scikit-learn, LangChain, pandas, NumPy.

WORK EXPERIENCE

Graduate Researcher | University of Cincinnati Feb 2026 – Present

- Designed two-stage memory-augmented LLM pipeline with agentic tool calling for Android GUI bug reproduction, benchmarked against a published GPT-4o vision baseline across 53 runs on 20+ real-world apps.
- Applied prompt engineering across a provider-agnostic LLM abstraction (Gemini, GPT-4o, Llama, Qwen) and demonstrated **25.7x latency reduction** (39.4s vs 1012.1s avg per run) while preserving task-completion fidelity.

Software Development Engineer | Siemens Industry Software Jul 2022 – Jun 2024

- Automated **Teamcenter X** infrastructure workflows with Python (Boto3) and REST APIs, replacing legacy Ruby scripts and reducing deployment time by **40%** across AWS cloud environments.
- Debugged and resolved live production issues across **44 environment upgrades** in 4 months, profiling and optimizing Linux, EC2, database, and Teamcenter X service performance.
- Designed **Ansible-based Blue-Green deployment** framework, reducing production upgrade downtime by **50%** and enabling scalable, repeatable customer environment rollouts.
- Migrated **40+ production environments** to Kubernetes using ArgoCD, Helm, GitLab CI/CD, and Git-based version control, modernizing deployment architecture at scale.
- Served as Quality Coach and SonarQube Specialist, raising automated test coverage from **35% to 78%** and mentoring **5 engineers** through onboarding.

Software Developer Intern | Siemens Industry Software Mar 2022 – Jun 2022

- Automated provisioning of AWS cloud resources with **Terraform**, reducing deployment time by **30%**.
- Developed **Go + Terratest** validation framework, catching **95%** of misconfigurations pre-production and preventing faulty infrastructure changes from reaching customer environments.

Software Developer Intern | Government of Maharashtra, Water Resource Department Sep 2021 – Dec 2021

- Built and deployed a full-stack **React and MySQL** workflow application replacing paper-based departmental record management, cutting manual record-processing effort by **40%**.

PROJECTS AND RESEARCH

PolicyPilot – Medical Benefit Policy Intelligence Portal [React, TS, Node, Gemini, MCP] May 2026

- Architected evidence-grounded **RAG pipeline** over Gemini 2.5-flash and designed a **17-tool MCP server** exposing structured LLM tool-calling for PA extraction, consumable by external AI agents.

EXTRACURRICULAR ACTIVITIES

- **Finalist, RevolutionUC 2025 Hackathon** (300 participants, 31 schools) | **SAFe Practitioner 5.1** (Mar 2023).